David Danielian

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SUMMARY

Full-stack and Cloud Infrastructure Engineer with experience in GPU-accelerated pipelines, web development, robotics, and spatial reconstruction

WORK EXPERIENCE

Smartapp.com Worcester, MA Sept 2023 - Present

Cloud and DevOps Engineer

- Developed and deployed a scalable, containerized computer vision pipeline leveraging COLMAP and 3D Gaussian Splatting to create spatial reconstructions of indoor spaces from video input.
- Architected pipeline end to end, including configurable step-based Python code, designing SQL and No-SQL databases, Docker containerization, and job orchestration via Kubernetes and Argo Workflows.
- Set up CI/CD workflows including automated test suites, and automated Docker image builds and cloud function deployment via Jenkins, allowing for easy weekly releases.
- Built dashboards to monitor per-job costs and pipeline metrics in real-time.
- Configured GKE clusters, cross-project IAM, Workload Identity, API Secret Management, and Firestore-based distributed locking to support reliable concurrent pipeline execution.

Catchpoint Remote May 2022 - Aug 2022

Software Engineering Intern

- Updated legacy web pages written in .NET Web Forms to React GraphQL stack, including documenting functionality of the legacy pages to scope required changes

Human-Computer Interaction Lab WPI Worcester, MA May 2021 - May 2023

Research Software Developer

- Built web-based fNIRS signal processing and analysis tools for neuroscience researchers
- Implemented queueing and websockets to enable the preprocessing of large datasets

PROJECTS

Brain-Computer Interface to Improve Educational Software

WPI Capstone project. Developed a Brain-Computer interface that uses ML models to monitor and visualize a user's cognitive state while doing learning tasks. Created Chrome extension with Flask API to receive brain data and display visuals in the browser. Also conducted user studies to evaluate efficacy of features that used brain data to intervene and assist users in learning tasks.

VR Haptic Gloves

Developed VR haptics gloves with finger position tracking, and force feedback. Responsible for sensor analysis, writing firmware, handling IPC, and creating test environments in Unity**.**

SKILLS

Cloud & Infrastructure: GCP, Docker, Kubernetes, Argo Workflows, Firestore, Cloud Functions, AWS, Lambda
Web Development: HTML, CSS, Next.js, React, Vue, Flask, Three.js, Stripe, Firebase
Programming Languages: Python, JavaScript/TypeScript, C#, SQL, Java, C++, C
Tools: Git, Linux, Jenkins, ROS, Unity, Godot, Figma

EDUCATION

Worcester Polytechnic Institute Worcester MA **Class of 2023**

Bachelors in Computer Science, Robotics Engineering Minor, 3.76 GPA

References are available upon request to protect my mentors and colleagues privacy